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RESEARCH ARTICLE

Methane and nitrous oxide emissions during biochar-composting are driven by biochar application rate and aggregate formation

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Abstract

Manure is a leading source of methane (CH_4), nitrous oxide (N_2O), and ammonia (NH_3) emissions, and alternative manure management practices can help society meet climate goals and mitigate air pollution. Recent studies show that biochar-composting can substantially reduce emissions from manure. However, most studies test only one type of biochar applied at a single application rate, leading to high variation in emission reductions between studies. Here, we measured greenhouse gas and NH_3 emissions during biochar-composting of dairy manure with biochar applied at 5% or 20%, by mass, and made from walnut shells, almond shells, or almond clippings. We found little difference in emissions between biochar type. However, we found that the 20% application rates increased CH_4 emissions and decreased N_2O and NH_3 emissions, resulting in a net reduction in global warming potential (GWP). We attribute this result to biochar increasing the formation of compost aggregates, which likely acted as anaerobic reactors for methanogenesis and complete denitrification. Biochar may have further fueled CH_4 production and N_2O consumption by acting as an electron shuttle within aggregates. We recommend lower application rates, as we found that the 5% treatments in our study led to a similar reduction in GWP without increasing CH_4 emissions.

KEYWORDS

ammonia, biochar, climate change mitigation, composting, livestock, manure, methane, nitrous oxide

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1 | INTRODUCTION

Manure management is a leading source of anthropogenic methane (CH_4), nitrous oxide (N_2O), and ammonia (NH_3) emissions. In the United States, manure management accounts for 9% of CH_4 emissions and 4% of N_2O emissions, while in California, the nation's leader in dairy production, manure contributes 26% and 12% of statewide CH_4 and N_2O emissions, respectively (CARB, 2022a, 2022b; U.S. EPA, 2023). Because the 100-year global warming potential (GWP) of CH_4 and N_2O are 28 and 264 times that of carbon dioxide (CO_2), respectively, reducing the emission of these gases from manure is a low-hanging fruit when it comes to climate change mitigation and can help society make significant progress on rapidly approaching climate goals (Crippa et al., 2021; Herrero et al., 2016; IPCC, 2023; Rojas-Downing et al., 2017).

Recent efforts to curb emissions from livestock manure have focused on the use of anaerobic digesters, which trap CH_4 emissions from manure lagoons for use in energy production (CARB, 2022c; Edwards et al., 2015; Satchwell et al., 2018). While this strategy can lead to large reductions in CH_4 emissions from liquid manure, solid manure that is separated from digesters is often stockpiled and can remain a significant source of both CH_4 and N_2O (Owen & Silver, 2015; Pattey et al., 2005). Aerobically composting these manure solids, rather than stockpiling them, can reduce CH_4 emissions while producing a valuable organic fertilizer (Hou et al., 2015; Pardo et al., 2015). However, composting nitrogen (N) rich manure can lead to high N_2O emissions, which risks reducing the climate benefit of composting by essentially "pollution swapping" CH_4 for N_2O (McNicol et al., 2020; Pardo et al., 2015; Pattey et al., 2005). Furthermore, manure composting can result in large losses of reactive N to the environment through the volatilization of ammonia (NH_3), an odorous gas and an important precursor to $PM_{2.5}$ pollution (Nordahl et al., 2020; Nordahl et al., 2023). In order to improve the effectiveness of manure composting as a climate change mitigation strategy, novel composting techniques are needed that can minimize the emission of greenhouse gases (GHGs) and NH_3 from solid manure while transforming this polluting waste into a valuable organic resource.

Biochar-composting is one strategy that has been recently proposed for reducing the emission of CH_4, N_2O, and NH_3 from manure (Akdeniz, 2019; Sánchez-Monedero et al., 2018; Steiner et al., 2015). While the technique of combining charcoal and organic waste to improve soil fertility has been practiced by multiple Indigenous Peoples for millennia, biochar-composting for emission mitigation is a novel manure management strategy that has only recently been investigated (Glaser et al., 2001; Glaser & Birk, 2012; Ogawa & Okimori, 2010; Sánchez-Monedero et al., 2018). Several studies suggest that the high porosity of biochar can improve compost pile aeration and help prevent the formation of anaerobic microsites where CH_4 is produced (Agyarko-Mintah et al., 2017; Awasthi et al., 2016; Sonoki et al., 2013; Vandecasteele et al., 2016). In a recent study, we found that biochar-composting reduced CH_4 emissions by 84% relative to composting without biochar, resulting in a climate-negative dairy manure management strategy (Harrison et al., 2022). Biochar has also been found to substantially cut N_2O and NH_3 emissions from soils and composts, likely by adsorbing precursors to these gases on the surface of biochar and by promoting N_2O reduction to N_2 through the shuttling of electrons to denitrifiers (Cayuela et al., 2013; Hua et al., 2009; Steiner et al., 2010; Wang et al., 2013).

Despite the promising potential of biochar to reduce emissions during composting, most biochar-composting studies have tested only one type of biochar applied at a single application rate, leading to a wide range in reported emission reductions between studies (Akdeniz, 2019; Harrison et al., 2022; Sánchez-Monedero et al., 2018). Biochar can be made from essentially any organic material, and it retains many of the original properties of the feedstock from which it is made from. Because of this, key physical and chemical characteristics of biochars such as porosity, surface area, and functionality, can vary greatly depending on their feedstock, and this can influence a particular biochar's capacity to reduce emissions (Pascual et al., 2020; Sánchez-Monedero et al., 2018; Xiao et al., 2017). Furthermore, different application rates of biochar to manure could lead to a wide range of composting environments, differing in physical structure, redox conditions, and nutrient compositions, all of which can lead to dramatic changes in the composting process and its associated emissions (Akdeniz, 2019; Hua et al., 2009; Sánchez-Monedero et al., 2018). Therefore, an investigation into the optimal biochar feedstocks and application rates for emission mitigation during composting, as well as the mechanisms driving these potential reductions, is desperately needed to provide farmers with specific recommendations for biochar-composting. Testing biochars made from locally abundant sources of organic resources is especially needed in order to strategically harness biomass streams in agricultural regions to help develop circular, climate-smart economies (Nematian et al., 2021).

Here, we conducted a laboratory-scale biochar-composting experiment to investigate the optimal biochar feedstocks and application rates for mitigating GHG and NH_3 emissions from dairy manure. We tested biochars made from either walnut shells, almond shells, or almond clippings and applied biochar at a rate of 5% or 20% by mass. We hypothesized that higher application rates of biochar would result in the greatest reductions in emissions

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and that biochars with the highest porosity would be most effective in mitigating CH_4 emissions through enhanced pile aeration. We also hypothesized that biochars with the highest surface area would result in the greatest reduction in N_2O and NH_3 emissions, through the absorption of gas precursors and NH_3 on the biochar surface.

2 | METHODS

2.1 | Compost experiment design

We conducted a composting experiment in aerated, insulated bioreactors over 6 weeks in the laboratory, which was the time it took for the compost to reach maturity and meet pathogen requirements (see Section 3.1). Our experimental design consisted of a no-biochar control and six biochar treatments, each tested in triplicate. Treatments consisted of biochars applied at a rate of 5% or 20% by dry mass, and produced from either almond shells, almond clippings, or walnut shells. These feedstocks were used because orchard biomass has been identified as a large source of available feedstock for biochar production in California's Central Valley where the majority of the state's dairies are also located (Marklein et al., 2021; Nematian et al., 2021). Biochars were produced at approximately 350°C in a mobile, continuous feed pyrolizer (AC Fox, Incorporated, Groton, MA, USA).

Compost bioreactors were constructed from a polyethylene drum (38.1 cm inner diameter, 53.4 cm height, 0.06 m^3 volume) and covered with a removable, air-tight lid (Figure 1a). Each drum was insulated with 7.62 cm of polyurethane foam to ensure that thermophilic composting temperatures were met (Steiner et al., 2010). Each experimental block had its own compressed air source, which was controlled by a pressure regulator, while the volume of air flowing into each bioreactor was regulated with a mechanical flowmeter set at a rate of 0.6 L min^{-1} (0.5 L min^{-1} $kg\,VS^{-1}$) (Li et al., 2008; Steiner et al., 2010). Air was dispersed at the bottom of each reactor through a coiled, 167 cm long tube, with 20 evenly spaced 4 mm wide holes. The compost feedstock sat 7.62 cm above the tube on a perforated steel plate. Out gas was allowed to pass out of the top of the bioreactor through a 5.1 cm wide hole in the middle of the air-tight lid.

Fresh solid dairy manure was obtained from a solids separator at a local dairy farm the day before the experiment began. Each treatment and the control received 15 kg of fresh dairy manure, and the 5% and 20% treatments also received 0.17 and 0.83 kg of biochar, respectively. Due to the very low moisture content of the biochars, the 5% and 20% treatments also received 0.6 and 2.9 kg of deionized water, respectively, to bring the overall gravimetric moisture content of the manure and biochar mixtures up to approximately 74%, the same as the no-biochar control. Before the feedstocks were loaded in the bioreactor, the

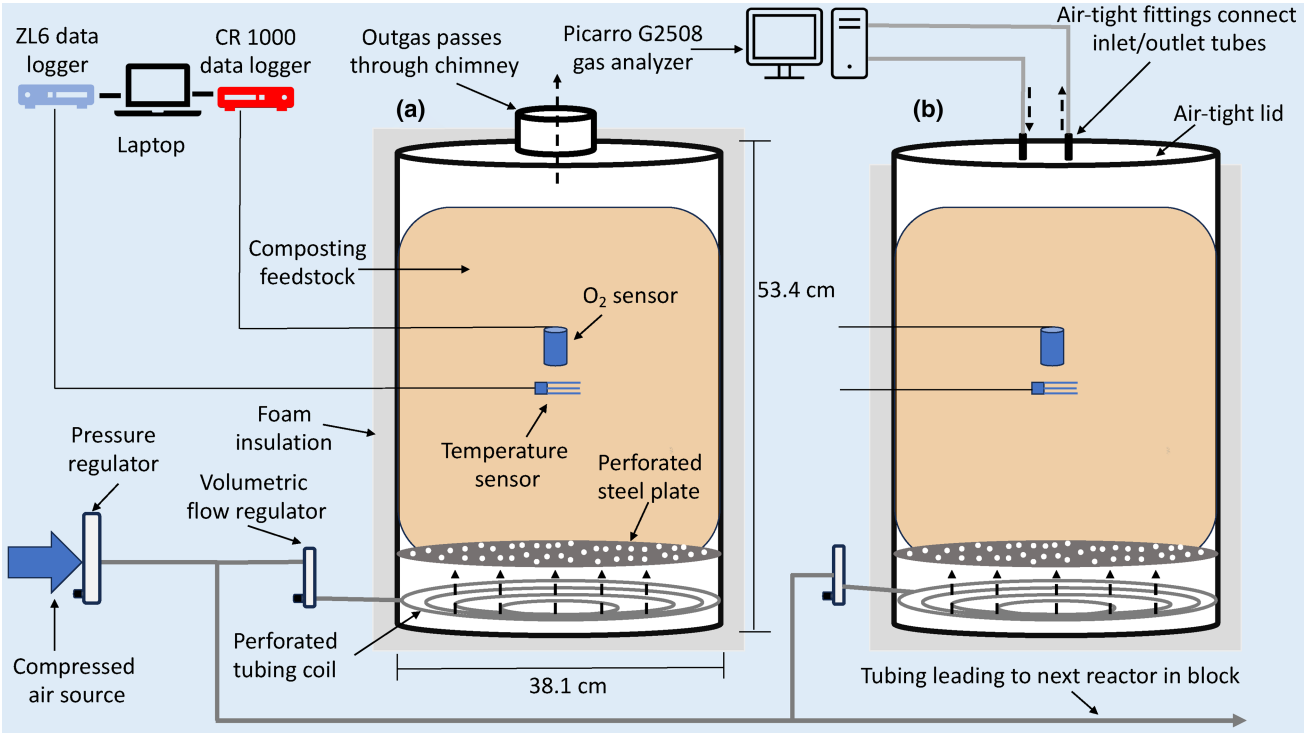


FIGURE 1 Compost reactor design showing sensor set up and aeration system of (a) a reactor that is not being measured for gas emissions and (b) a reactor being measured for gas emissions. Dotted arrows indicate the direction of airflow.

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contents of each treatment and the no-biochar control were mixed thoroughly in a mechanical mixer, lined with a clean plastic bag, for 3 min at a rate of 36 rotations per minute (rpm). Similarly, composting feedstocks were mixed weekly by emptying the contents of each bioreactor into the lined mixer and rotating for 1 min at 36 rpm. Each bioreactor was assigned to a random position within each block, and new positions were randomly assigned after compost mixing each week.

2.2 | Compost and biochar characterization

In one of the three experimental blocks, each bioreactor was equipped with an oxygen sensor (Apogee SO-110; Apogee Instruments, Logan, UT, USA) connected to a data logger (CR 1000; Campbell Scientific, Logan, UT, USA) and a temperature sensor (5TE; Campbell Scientific) connected to a separate data logger (ZL6 Basic; Meter Group, Pullman, WA, USA) (Figure 1a). Oxygen sensors were calibrated before the experiment and weekly during the experiment according to the manufacturer's instructions. Sensor data were logged hourly, and daily averages are reported.

Each week, compost samples were taken after homogenization through mechanical mixing. Compost moisture was measured through oven drying at 55°C for 72 h and total carbon (C) and nitrogen (N) were measured from the dried samples on an elemental analyzer (Costech 4010; Costech Analytical Technologies, Inc., Valencia, CA, USA) in the Stable Isotope Ecosystem Laboratory of UC Merced. Compost ammonium (NH_4^+-N) and nitrate (NO_3^--N) were extracted in 2 M KCl and analyzed on a microplate reader (Synergy HTX Multimode Reader; Agilent Technologies, Inc., Santa Clara, CA, USA) through colorimetry. Compost dissolved organic carbon (DOC) was extracted in 0.5 M K_2SO_4, filtered through a 0.45 μm syringe filter, and measured on a TOC analyzer (Shimadzu TOC-Vcsh Total Organic Carbon Analyzer; Shimadzu Scientific Instruments, Kyoto, Japan). The pH and electrical conductivity (EC) of the composts were measured in a 1:5 (w/v) compost to DI water suspension. Volatile solids and ash contents of the composts and feedstocks were determined through proximate analysis according to ASTM D3172-13 (ASTM, 2021). Maturity of the final compost was tested by determining the germination index of each sample according to Luo et al. (2018) (Table S1). Aggregates in the final composts were measured by first sifting the contents of each bioreactor through a 25.4 mm screen, and then by sifting the material that passed through the first screen through a second, 6.35 mm screen. The mass of compost aggregates in each of the three size fractions (>25.4 mm, 6.35–25.4 mm, and <6.35 mm) was then measured for each bioreactor (Table S1).

Each of the three biochar types were also analyzed for total C and N, NH_4^+-N and NO_3^--N, pH and EC, and volatile solids and ash content using the same methods used for the composts (Table S2). The total O and H of each biochar were determined using a thermal conversion elemental analyzer (TC/EA High Temperature Conversion Elemental Analyzer; Thermo Scientific, Waltham, MA, USA). Each biochar's specific surface area and pore size were also measured on a TriStar II Plus (Micromeritics, Norcross, GA, USA) using the Brunauer, Emmett, and Teller method (Bardestani et al., 2019).

2.3 | Gas flux measurements

Greenhouse gases (CO_2, CH_4, and N_2O) and NH_3 were measured daily for the first 16 days, when the composting conditions were most dynamic, and then approximately every other day for the 42-day experiment. All gases were measured using a closed-circuit, static chamber connected to a cavity ring-down laser spectrometer (Picarro G2508; Picarro, Inc., Santa Clara, CA, USA). A chamber was created by removing the lid of the bioreactor and replacing it with a separate, air-tight lid with two 1/4″ tube fittings connected to the inlet and outlet ports of the gas analyzer with FEP tubing (Figure 1b). Gas was sampled from each reactor for 5 min and fluxes were calculated in the Picarro Soil Flux Processor program using a linear model or the Hutchinson and Mosier exponential model when nonlinear changes in concentration were observed (Hutchinson & Mosier, 1981). As the composting process proceeded, the headspace above the compost increased as mass was lost, and this change in headspace was accounted for before each flux measurement and calculation. Cumulative emissions were estimated by calculating the area under each flux curve using the trapezoid rule. We used 100-year GWP's (28 and 264 for CH_4 and N_2O, respectively) to convert cumulative CH_4 and N_2O to carbon dioxide equivalents (CO_2e) (IPCC, 2023).

2.4 | Data analysis

All statistical tests were performed in R version 4.2.3 (R Core Team, 2020). Differences in compost characteristics between treatments over the course of the experiment were analyzed using a two-way mixed analysis of variance (ANOVA), with application rate (0%, 5%, or 20%) and amendment type (none, walnut shell, almond shell, or almond clippings), and their interaction as fixed effects and block and time as random effects. Pairwise comparisons of means using Tukey's

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HSD test were then performed to test for significant differences between treatments. Two-way ANOVA and Tukey's HSD post hoc tests were used to test whether differences in cumulative gas emissions between application rate, biochar type, and their interaction, were significant. Significance for all statistical analyses was set at $p < 0.05$. Data was checked for normality and homogeneity before conducting all statistical analyses, and non-normal data was transformed using Box–Cox transformations (Box & Cox, 1964). Linear regression was also used to investigate the relationship between compost properties and gas emissions.

3 | RESULTS

3.1 | Compost physical and chemical characteristics

The gravimetric moisture contents of the composts slightly increased over the course of the experiment, or remained nearly constant, and there was no significant difference between amendment type or application rate (Figure 2c). Compost temperatures peaked on Days 2 and 3 of the experiment, and maximum temperatures were consistent, ranging from 66 to 68°C (Figure 2a). All composts were at thermophilic temperatures (>55°C) for 3 days, which meets the USDA requirement for pathogen control during thermophilic composting (NOSB, 2006). The mesophilic stage (<55°C) was reached by all composts by Day 5, and temperatures cooled to slightly above ambient temperatures shortly thereafter, where they remained for the duration of the experiment. The oxygen concentration for all composts at the start of the experiment was below 10%, but all biochar treatments reached ideal aerobic composting conditions (15%–20%) by Day 3 (Figure 2b) (Bernal et al., 2009). However, the no-biochar treatment remained at very low oxygen concentrations (<4%) until Day 7.

Total C concentrations remained stable over the course of the experiment for all composts while total N concentrations increased, and there was a significant difference in C between amendment type ($p < 0.001$), application rate ($p < 0.001$), and their interaction ($p < 0.001$) and a significant difference in N between amendment type ($p < 0.001$), application rate ($p < 0.001$), but not their interaction ($p = 0.105$) (Figure 2d,e). All biochar treatments had significantly higher concentrations of C relative to the no-biochar control, and all 20% biochar application treatments had significantly lower N concentrations compared with the 5% application treatments and the no-biochar control. There was no significant difference in NO_3^- or NH_4^+ content between treatments, which both declined as compost matured (Figure 2f,g). DOC content peaked in Week 2 and declined thereafter, and there was a significant difference between amendment type ($p < 0.001$), application rate ($p < 0.001$), and their interaction ($p < 0.001$), with almond clippings 20% and walnut shell 20% having lower DOC concentrations than the other biochar treatments and the no-biochar control (Figure 2h). We also observed a significant difference in pH between amendment type ($p < 0.01$) as the almond shell 20% treatment had significantly higher pH compared with the no-biochar control (Figure 2i).

At the end of the experiment, the final composts were sifted to determine the fraction of compost mass in aggregates (Table S1). There was a significant difference between application rates in the fraction of compost mass in total aggregates (>6.5 mm) ($p < 0.001$), medium aggregates (6.5–25.4 mm) ($p < 0.01$), and large aggregates (>25.4 mm) ($p < 0.05$). The 20% biochar application treatments had a significantly greater proportion of their mass in total and medium aggregates than the 5% and no-biochar control, as well as significantly more large aggregates than the no-biochar control. Differences in aggregate mass between the 5% treatments and the control were not statistically significant for any of the size fractions. The final composts were determined to be mature by the end of the experiment as they had an NH_4^+ content below 400 mg kg^{-1}, a DOC content below 4 g kg^{-1}, and a germination index above 50 (Table S1) (Bernal et al., 2009).

3.2 | GHG and ammonia emissions

A peak in CO_2 emissions occurred for all treatments and the control during the thermophilic stage when compost temperatures were highest, and there was a significant difference in CO_2 emissions between application rate ($p < 0.001$) (Figure 3a,e). The 5% ($p < 0.05$) and 20% ($p < 0.001$) biochar application treatments significantly reduced cumulative CO_2 emissions relative to the no-biochar control (Table 1). There was also a significant difference in cumulative CO_2 emissions between amendment type ($p < 0.001$) and the interaction between amendment type and application rate ($p < 0.05$) (Table S3). Almond shell biochar treatments had significantly higher CO_2 emissions than almond clippings biochar ($p < 0.01$) and walnut shell biochar treatments ($p < 0.01$), but there was no significant difference between almond shell biochar and no biochar ($p = 0.119$).

CH_4 fluxes from all composts peaked during the thermophilic stage of composting (Figure 3b,f). There was no significant difference in cumulative CH_4 emissions between amendment types ($p = 0.721$), but there was a significant difference between application rates ($p = 1.56 \times 10^{-9}$). While there was no significant difference between the no-biochar control and the 5% application treatments ($p = 0.972$), the 20% application treatments significantly increased CH_4 emissions relative to the control ($p < 0.001$) and the 5% treatments ($p < 0.001$), by an order of magnitude (Table 1).

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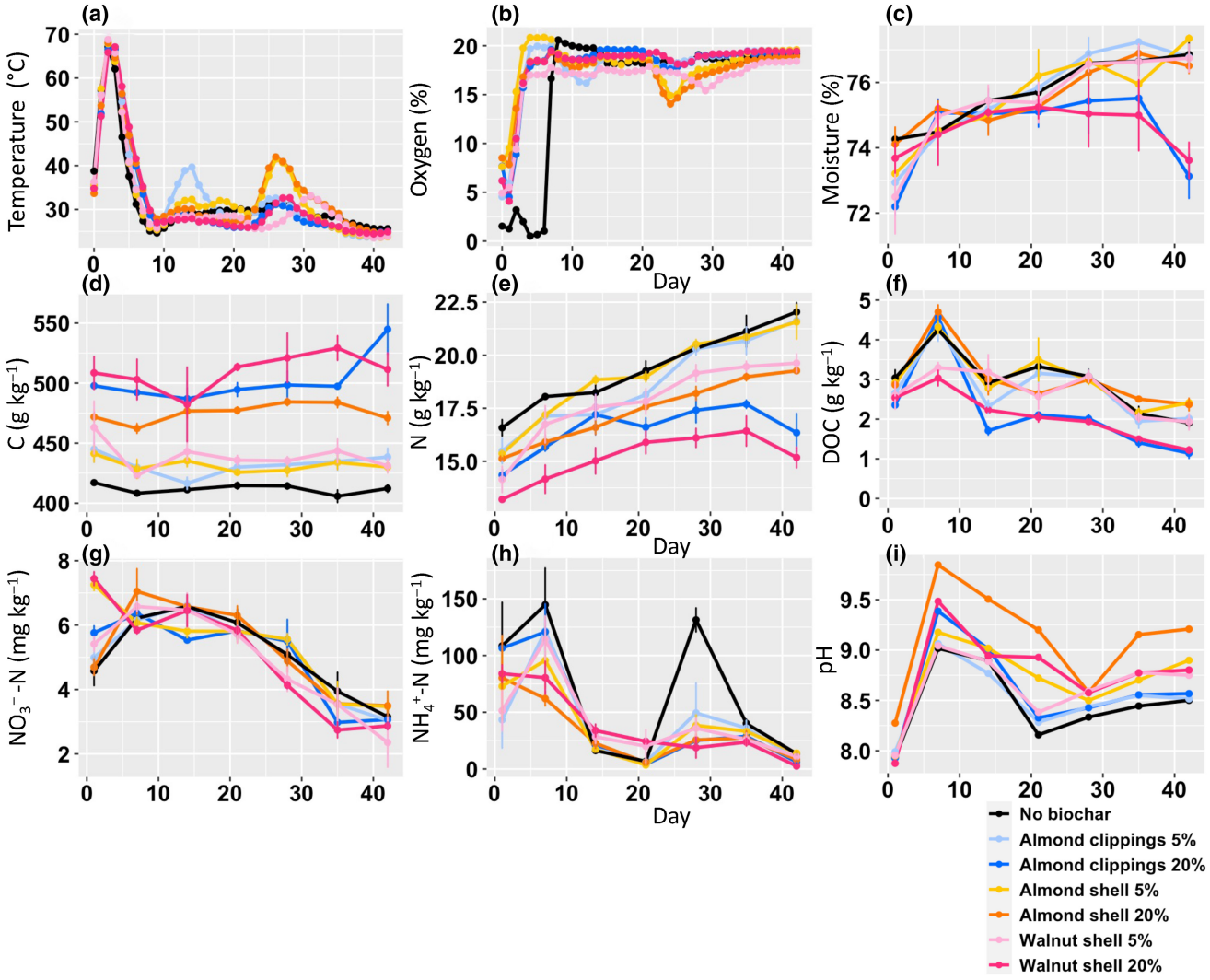


FIGURE 2 Physical and chemical properties of each biochar-compost treatment and the control over the course of the experiment showing (a) temperature in °C, (b) percent oxygen concentration, (c) percent gravimetric moisture, (d) C in g kg^{-1}, (e) N in g kg^{-1}, (f) DOC in g kg^{-1}, (g) NO_3^--N in mg kg^{-1}, (h) NH_4^+-N in mg kg^{-1} and (i) pH. Error bars are ±1 standard error. DOC, dissolved organic carbon.

Nearly all N_2O and NH_3 were emitted on Day 2 of the experiment (Figure 3c,d,g,h). There was no significant difference in cumulative N_2O ($p=0.949$) or NH_3 ($p=0.735$) emissions between amendment type. However, the difference in N_2O ($p<0.01$) and NH_3 ($p<0.01$) between application rates was significant (Table 1). Though not statistically significant, the 5% biochar application treatments, on average, reduced N_2O ($p=0.0895$) and NH_3 ($p=0.354$) emissions by 64% and 33%, relative to the no-biochar control, respectively. The 20% application treatments, on average, significantly reduced N_2O ($p<0.01$) by 84% and NH_3 ($p<0.01$) by 65%, relative to the control, respectively.

Differences in net 100-year GWP between amendment types were not statistically significant ($p=0.944$). However, biochar application rate had a significant effect on GWP ($p<0.01$) (Table 1). The 5% application treatments significantly reduced GWP by 63%, on average, relative to the no-biochar control ($p<0.05$), while the 20% application treatments reduced GWP by 70%, on average, relative to the control ($p<0.01$). Differences in GWP between the 5% and 20% application treatments were not significant ($p=0.874$).

4 | DISCUSSION

4.1 | GHG and ammonia emissions

4.1.1 | CO_2 emissions

CO_2 emissions released during composting are biogenic in origin and are a part of the fast carbon cycle, so they do not have an overall impact on the climate (Pardo et al., 2015). However, CO_2 emissions indicate the rate of aerobic decomposition, and they are useful in determining the

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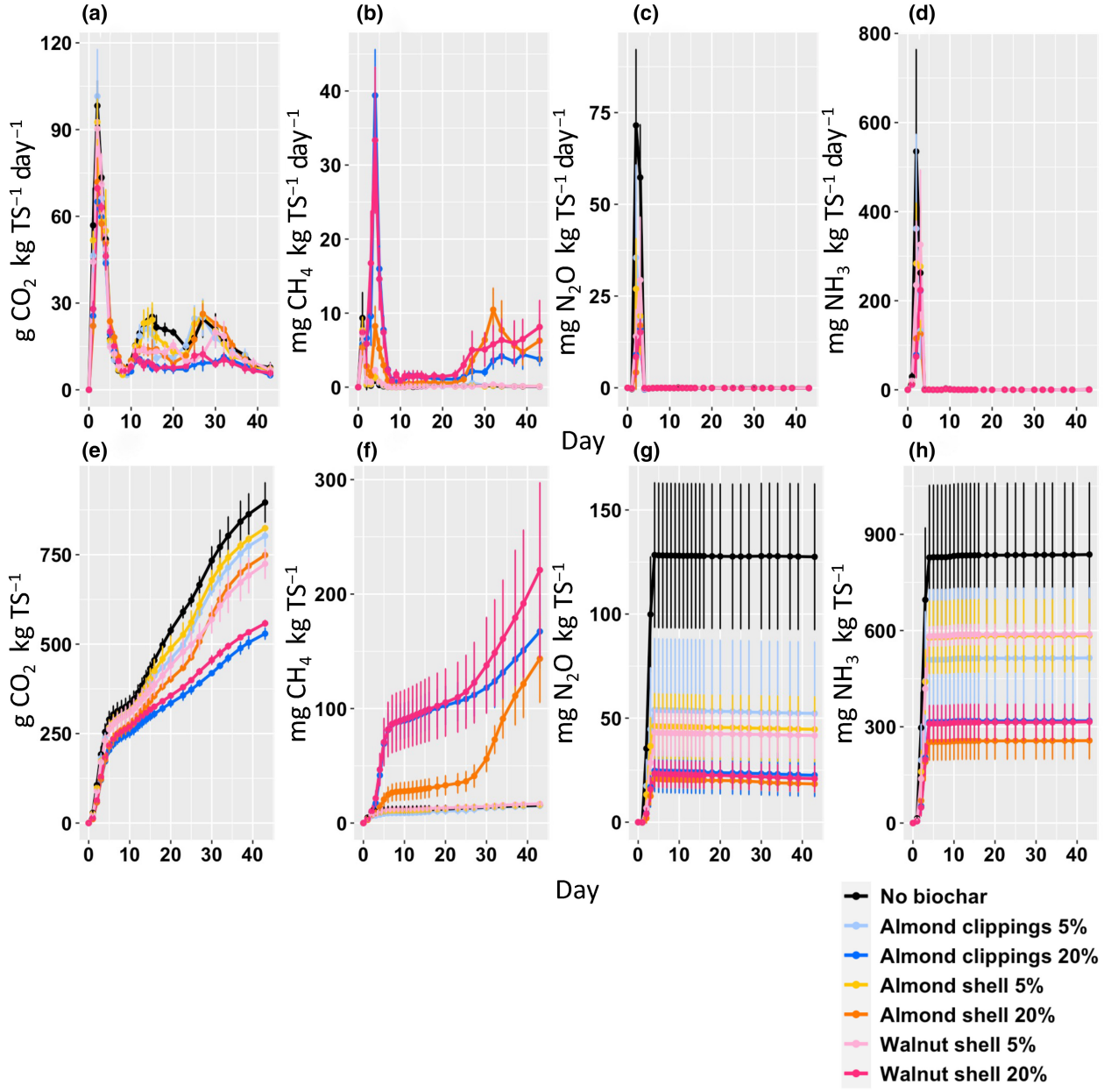


FIGURE 3 Daily average flux emission of (a) CO_2 in $g\,CO_2\,kg\,TS^{-1}\,day^{-1}$, (b) CH_4 in $mg\,CH_4\,kg\,TS^{-1}\,day^{-1}$, (c) N_2O in $mg\,N_2O\,kg\,TS^{-1}\,day^{-1}$, (d) NH_3 in $mg\,NH_3\,kg\,TS^{-1}\,day^{-1}$ and cumulative emission of (e) CO_2 in $g\,CO_2\,kg\,TS^{-1}$, (f) CH_4 in mg CH_4 kg TS^{-1}, (g) N_2O in $mg\,N_2O\,kg\,TS^{-1}$ and (h) NH_3 in $mg\,NH_3\,kg\,TS^{-1}$. Error bars are ±1 standard error.

TABLE 1 Mean (±standard error) cumulative CO_2, CH_4, N_2O NH_3, and 100-year GWP for treatments grouped by application rate.

	0% biochar (control)	5% biochar	20% biochar
Cumulative CO_2 ($g\,CO_2\,kg\,TS^{-1}$)	896 ± 55.3 a	784 ± 21.3 b	612 ± 35.1 c
Cumulative CH_4 ($mg\,CH_4\,kg\,TS^{-1}$)	15.4 ± 2.13 a	15.9 ± 1.80 a	177 ± 28.7 b
Cumulative N_2O ($mg\,N_2O\,kg\,TS^{-1}$)	127 ± 35.1 a	46.2 ± 11.5 ab	20.6 ± 3.68 b
Cumulative NH_3 ($mg\,NH_3\,kg\,TS^{-1}$)	837 ± 224 a	563 ± 72.6 a	297 ± 25.6 b
GWP ($g\,CO_2e\,kg\,TS^{-1}$)	34.2 ± 9.32 a	12.7 ± 3.04 b	10.4 ± 4.07 b

Note: The 0% application rate is the same as the no-biochar control. Letters denote statistical significance between application rates ($p < 0.05$).

Abbreviation: GWP, global warming potential.

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stability of the final compost produced (Bernal et al., 2009). While the overall trend in CO_2 emissions was consistent between treatments and the control, we found that all biochar treatments significantly reduced CO_2 emissions. Some biochar-composting studies observed an increase in CO_2 emissions following the application of biochar, likely due to the capacity of biochar to increase microbial activity during composting through enhanced aeration (Czekała et al., 2016; Steiner et al., 2011). However, in a previous study, we also found that biochar suppressed CO_2 emissions during composting (Harrison et al., 2022). While all biochar treatments increased compost O_2 concentrations in our study, biochar may have reduced CO_2 fluxes by limiting respiration through the adsorption of labile C onto its surface (Lehmann et al., 2011). We find evidence for this hypothesis in the significantly lower DOC concentrations of the walnut shell 20% and almond clippings 20% treatments, which also had the lowest cumulative CO_2 emissions (Figures 2f and 3e). However, we did not find a significant correlation between CO_2 emissions and DOC (Figure S1).

4.1.2 | CH_4 emissions

In a previous field study, we found that a 6% (w/w) application rate of biochar to composting dairy manure reduced CH_4 emissions by 84% (Harrison et al., 2022). Surprisingly, we did not observe a significant difference in CH_4 emissions between the no-biochar control and the 5% biochar treatments in this study. Even more surprising, we found that the 20% biochar application treatments significantly increased CH_4 emissions by an order of magnitude. The no-biochar control and the 5% application treatments had a peak in CH_4 emissions on the first day of the experiment, when O_2 concentrations were low (Figures 2b and 3b). However, the 20% application treatments had a peak in CH_4 emissions on Day 3, when all biochar treatments had high O_2 concentrations, and the no-biochar control remained anaerobic. Despite increasing O_2 concentrations relative to the no-biochar control, the 20% biochar treatments increased CH_4 emissions.

This puzzling result led us to investigate whether biochar could be driving CH_4 emissions by providing a significant source of labile organic carbon, but differences in DOC between treatments were not statistically significant, except for the walnut shell 20% and almond clippings 20% treatments, which each had significantly lower DOC than the no-biochar control (Figure 2f). However, we did find a significant correlation between CH_4 emissions and the amount of compost mass in aggregates ($R^2 = 0.75, p < 0.001$) (Figure 4a,b). We hypothesize that biochar acted as a binding agent in the compost, enhancing aggregation through cation bridging between cations adsorbed to the biochar surface and the negatively charged compost (Figure 5). Previous analyses of the surface of co-composted biochars have found that composting enhances biochar surface oxidation and forms a mixed-charge organo-mineral layer on the surface of biochar (Archanjo et al., 2017; Hagemann et al., 2017; Kammann et al., 2015; Prost et al., 2013; Wiedner et al., 2015). The increase in biochar's reactivity through surface oxidation during composting may play an important role in enhancing biochar's ability to help form aggregates. Furthermore, the formation of an organo-mineral layer on the surface of biochar provides evidence for the capacity of biochar to adsorb composting organic matter through cation bridging.

While our sensors measured high O_2 concentrations in the bulk compost of the 20% biochar application treatments, the compost aggregates may have acted as anaerobic microsites that were undetectable by our bulk O_2 sensors. In soils, aggregates are known to be anaerobic hotspots, with O_2 diffusion to the core of aggregates limited by pore connectivity and microbial consumption (Keiluweit et al., 2017; Sexstone et al., 1985; Zausig et al., 1993). This anoxic environment allows for aggregates to function as biogeochemical reactors for methanogenesis and denitrification (Diba et al., 2011; Keiluweit et al., 2016; Kimura et al., 2012; Mangalassery et al., 2013; Sexstone et al., 1985; Wang et al., 2019). Compost aggregates, though largely unexplored, may serve a similar role in the production of CH_4 emissions during composting.

In addition to creating this anoxic environment, biochar may have also increased CH_4 emissions by acting as an electron donor for methanogenesis within aggregates (Figure 5). Biochars have been found to play a direct role in redox reactions through the transfer of electrons facilitated by (hydro)quinone functional groups and/or polyaromatic hydrocarbons (Klüpfel et al., 2014; Saquing et al., 2016). While previous studies have found that biochar can reduce CH_4 emissions through improved pile aeration when applied during aerobic composting, other studies have found that biochar can increase CH_4 emissions by mediating the transfer of electrons within anaerobic environments (Chen et al., 2014; Harrison et al., 2022; Pan et al., 2019; Sánchez-Monedero et al., 2018; Wang et al., 2020; Yuan et al., 2018). Aggregates in the 20% application treatments likely had more biochar available for electron shuttling than aggregates in the 5% application treatments. This may help explain why a relatively modest increase in aggregation from the 5% to the 20% application treatments led to a large difference in the amount of CH_4 emitted (Figure 4a,b).

It is also important to note that the average CH_4 emission factor for the 20% application treatments was still approximately an order of magnitude lower than that of the

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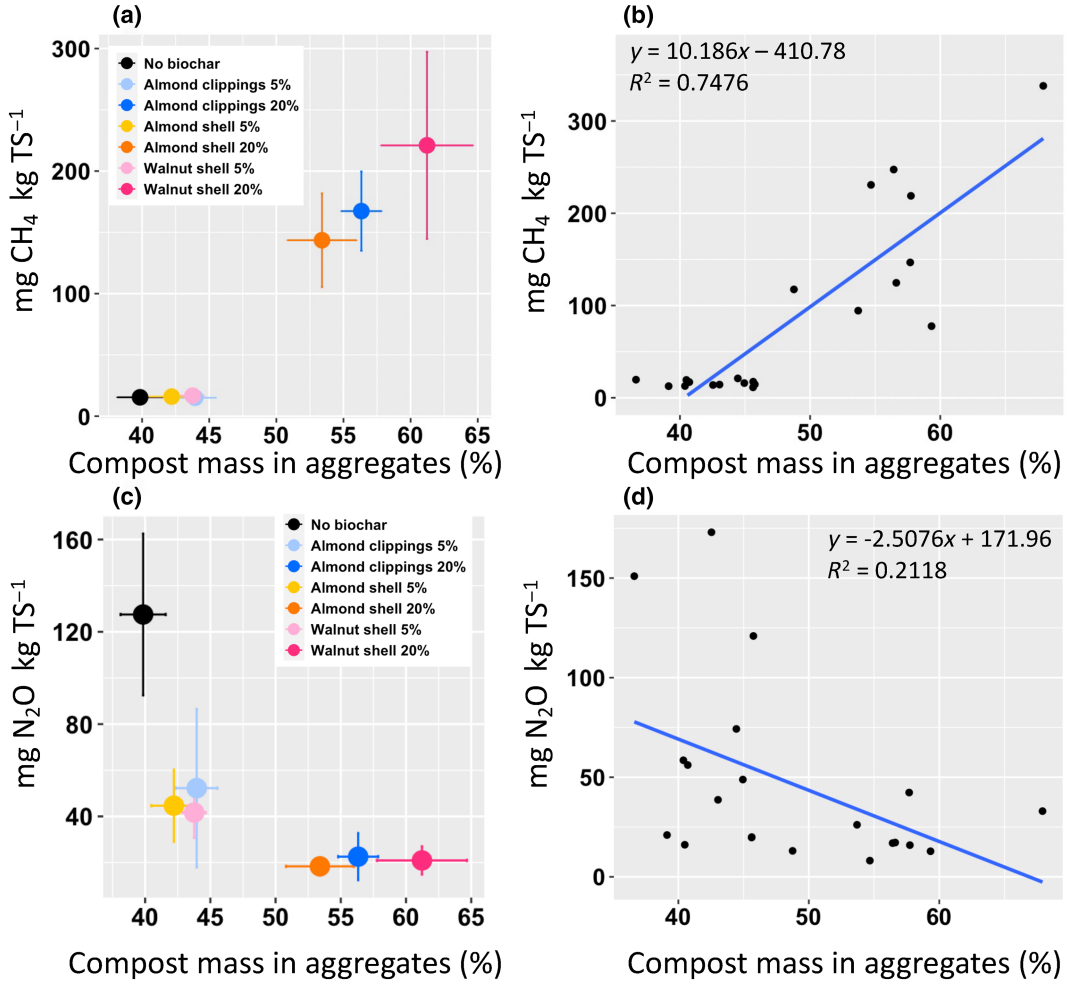


FIGURE 4 (a) Relationship between cumulative CH_4 emissions and the percent of compost mass in aggregates (>6.35 mm), grouped by treatment. Error bars are ±1 standard error. (b) Linear regression plot of cumulative CH_4 emissions and the percent of compost mass in aggregates (>6.35 mm). (c) Relationship between cumulative N_2O emissions and the percent of compost mass in aggregates (>6.35 mm), grouped by treatment. Error bars are ±1 standard error. (d) Linear regression plot of cumulative N_2O emissions and the percent of compost mass in aggregates (>6.35 mm).

compost windrows in our previous field study (Harrison et al., 2022). This shows the large variation in emissions between composting practices. For example, our in-vessel composting experiment, which had forced air flowing through the composting feedstock, likely facilitated a more aerobic environment than the large windrows from our field experiment. In composting conditions with limited airflow, such as windrows, the aeration effect by biochar may play a more important role, which may have led to the large reduction in CH_4 emissions from biochar-composting in the field experiment (Harrison et al., 2022). Different compost-turning practices may also help explain why we did not see a reduction in CH_4 emissions from the biochar-compost treatments in this study, as the slow rotating tumbler may have helped biochar form aggregates whereas the mechanical windrow turners used in the field study may have helped break up aggregates formed by biochar.

4.1.3 | N_2O emissions

Nearly all N_2O was emitted from the treatments and control on Days 2 and 3 of the experiment, when temperatures approached their maximum and O_2 concentrations were very low (Figures 2a,b and 3c). This suggests that N_2O was produced primarily through denitrification. We found that N_2O emissions were reduced on average by 64% and 84%, by the 5% and 20% biochar application treatments, respectively. In this anoxic environment, biochar likely reduced N_2O emissions by facilitating electron shuttling to promote complete denitrification to N_2 (Cayuela et al., 2013, 2014, 2015; Harter et al., 2014; Weldon et al., 2019). Biochar may also reduce N_2O emissions by slowing down the rate of denitrification through N immobilization, NO_3^- and DOC adsorption, and improved aeration, but we found no significant difference in NH_4^+ or NO_3^- concentrations, suggesting similar rates of denitrification (and nitrification) between

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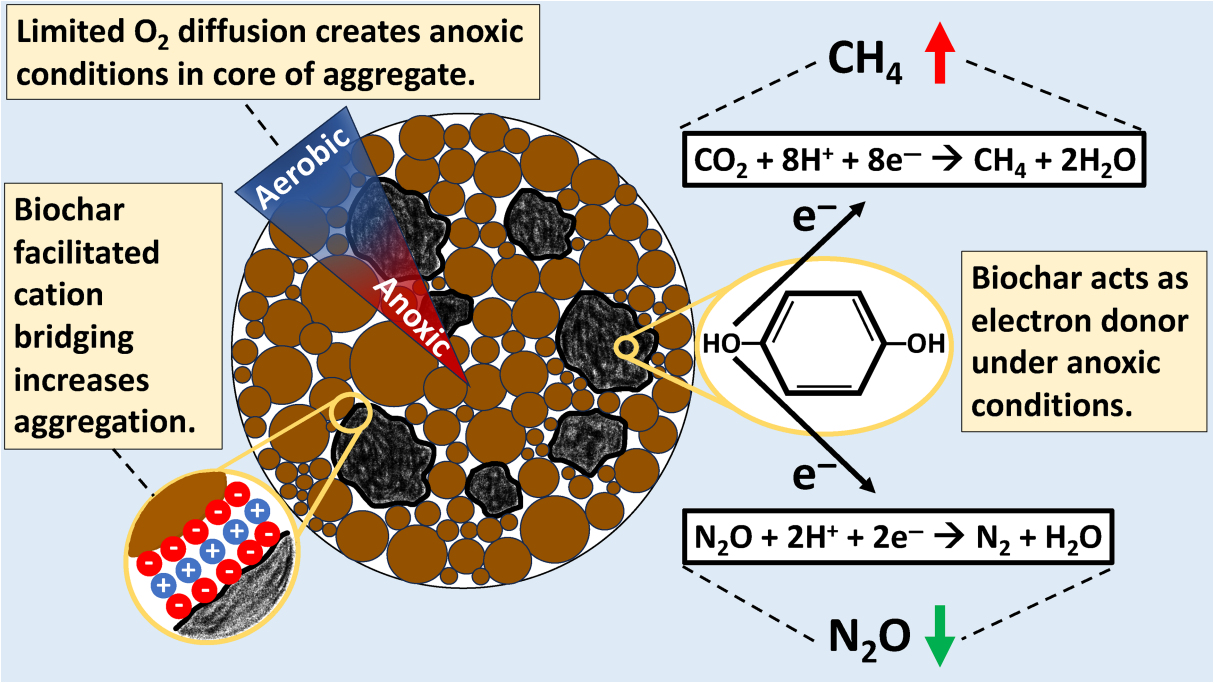


FIGURE 5 Conceptual model of the interaction between biochar and compost aggregates. Biochar increases aggregate formation through cation bridging. Aggregates likely have anoxic cores due to limited O_2 diffusion and rapid microbial consumption. In this anoxic environment, biochar pieces within aggregates likely drive CH_4 production and N_2O consumption by acting as an electron donor during methanogenesis and complete denitrification to N_2.

biochar treatments and the control (Cayuela et al., 2013; Harter et al., 2014). We also found a significant negative correlation between compost aggregates and N_2O emissions ($R^2 = 0.21$, $p < 0.05$) (Figure 4c,d). Aggregates can be hotspots for denitrification in soils (Diba et al., 2011; Keiluweit et al., 2016; Kimura et al., 2012; Sexstone et al., 1985; Wang et al., 2019). We found evidence that aggregation contributed to lower N_2O emissions, suggesting that aggregates may have functioned as reactors for complete denitrification, due to the very low O_2 concentrations at their core (Figure 5). Aggregates with biochar pieces in them may have further fueled N_2O consumption by promoting complete denitrification through electron shuttling (Figure 5).

4.1.4 | NH_3 emissions

Biochar's capacity to reduce NH_3 emissions from soils and composts through the adsorption of NH_4^+ and NH_3 has been well documented (Ghezzehei et al., 2014; Hua et al., 2009; Liu et al., 2019; Sánchez-Monedero et al., 2018; Spokas et al., 2012; Steiner et al., 2010; Taghizadeh-Toosi et al., 2012). In agreement with previous work, we found that the 5% and 20% biochar application treatments reduced NH_3 emissions by 33% and 65%, on average, respectively, although only the reduction by the 20% treatments was statistically significant (Table 1). Nearly all NH_3 emissions occurred in the first 3 days of composting for all treatments and the control, when temperatures were rising rapidly and O_2 concentrations were low (Figures 2a,b and 3d). This high rate of NH_3 volatilization is likely the result of an accumulation of NH_4^+ from rapid ammonification and low nitrification rates, which were likely inhibited due to high temperatures and low O_2 concentrations in the compost (Harrison et al., 2022; Vergara & Silver, 2019). Despite lower NH_3 emissions in the biochar treatments, there was no significant difference in NH_4^+ content during composting (Figure 2h). While we do not find direct evidence for the adsorption of NH_4^+ by biochar, it is possible that biochar reduced NH_3 emissions through the chemisorption of NH_4^+ through covalent bond formation. Chemisorbed NH_4^+ would be retained on the biochar surface and would explain why there was no difference in KCl-extractable NH_4^+ between biochar treatments and the control. Chemisorption has recently been shown to be an important and previously overlooked mechanism through which biochar can retain NH_4^+ (Hestrin et al., 2019; Krounbi et al., 2020, 2021).

4.2 | Optimizing biochar-composting for emission mitigation

It is interesting that differences in CH_4, N_2O, and NH_3 emissions between biochar type were not significant. This could be because differences in key biochar properties

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such as surface area, total pore volume, and pH were fairly consistent between biochar types and because biochar surface reactivity and redox potential, which likely acted as the primary controls on emissions in this experiment, are largely regulated by pyrolysis conditions (Cayuela et al., 2015; Harvey et al., 2012; Joseph et al., 2010; Klüpfel et al., 2014; Weldon et al., 2019). The low pyrolysis temperature of 350°C used to produce the biochars in this study likely led to similarities in reactivity and redox potential. These low temperature biochars each had high H:C and O:C ratios, which indicate a high proportion of oxygen-containing functional groups that are important for adsorption and electron shuttling by biochar (Table S2) (Klüpfel et al., 2014). In order to better understand which biochar properties are key to optimizing biochar-composting for emission mitigation, future research is needed that compares biochars with distinct properties. This could be done by producing biochars at varying temperatures or by comparing contrasting feedstocks, such as manure, wood, and grass. However, as mentioned previously, orchard biomass may be an ideal candidate for biochar production and use at California dairies due to the large amount produced in close proximity to dairies (Marklein et al., 2021; Nematian et al., 2021).

We did, however, find that biochar application rate has a significant impact on composting emissions. While the 5% application treatments did not affect CH_4 emissions, the 20% application treatments emitted over 10 times more CH_4 than the control. On the other hand, greater application rates led to a greater reduction in N_2O and NH_3 emissions. Due to the higher N_2O emission mitigation from the 20% application treatments, the average climate mitigation potential of the 5% and 20% treatments were similar, reducing net GWP by 63% and 70%, respectively (Table 1). However, higher application rates come at a higher financial cost to farmers and compost managers, and similar net reductions in emissions could be achieved at lower, more cost-effective application rates. Furthermore, we found that high application rates risk increasing CH_4 emissions. While the increase in CH_4 emissions was offset by the large reduction in N_2O emissions by the 20% application treatments in this study, this may not always be the case. Therefore, we recommend using lower biochar application rates to optimize the reduction in emissions during biochar-composting.

In this study, we found, for the first time, a strong relationship between compost aggregation and CH_4 emissions. In the composting literature, the term "anaerobic microsites" is often used to refer to the source of CH_4 emissions during aerobic composting (Harrison et al., 2022; Nordahl et al., 2023; Pardo et al., 2015; Sánchez-Monedero et al., 2018; Vergara & Silver, 2019). However, here we link CH_4 emissions to a physical property of compost piles that could be measured relatively easily by compost managers, which is a significant finding as we work toward optimizing manure composting for CH_4 mitigation. We propose that composting and biochar-composting should be managed to reduce aggregation in order to maximize aerobic conditions. This could be done by using turning practices that break up and destroy aggregates, such as mechanical windrow turners, and by avoiding turning methods that could promote aggregation, such as slow-rotating drums or tumblers for in-vessel composting. In the case of biochar-composting, lower biochar application rates may reduce the risk of excessive aggregate formation. Finally, using biochars produced at higher temperatures, which typically have low surface reactivity and greater aromatization, may also reduce the potential of biochar to bind with compost to help form aggregates, although, this could reduce the potential of biochar to adsorb gases, such as NH_3, and their precursors (Harvey et al., 2012).

In summary, we show that biochar-composting with low and high rates of biochar results in a similar reduction in net GWP during dairy manure composting, but high application rates risk "pollution swapping" N_2O emissions for CH_4 emissions. While we did not find a significant difference in emissions between biochar types, this was likely because the biochars tested had similar characteristics, and future research should test biochars produced at different temperatures and from contrasting feedstocks. We also found that aggregates formed during composting may act as anoxic biogeochemical reactors for methanogenesis and complete denitrification (Figure 5). Furthermore, the surface reactivity of biochar may help form aggregates while the electron shuttling capacity of biochar may drive redox reactions within aggregates. Because compost aggregates may be a significant source of CH_4 emissions, we recommend that compost managers take measures to reduce aggregate formation during manure composting.

AUTHOR CONTRIBUTIONS

Brendan P. Harrison: Conceptualization; formal analysis; investigation; methodology; visualization; writing – original draft. **Si Gao:** Investigation; writing – review and editing. **Touyee Thao:** Investigation; writing – review and editing. **Melinda L. Gonzales:** Investigation; writing – review and editing. **Kennedy L. Williams:** Investigation; writing – review and editing. **Natalie Scott:** Investigation; writing – review and editing. **Lauren Hale:** Supervision; writing – review and editing. **Teamrat Ghezzehei:** Conceptualization; supervision; writing – review and editing. **Gerardo Diaz:** Conceptualization; supervision; writing – review and editing. **Rebecca A. Ryals:** Conceptualization; supervision; writing – review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare they have no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in Dryad at https://doi.org/10.5061/dryad.905qfttsd.

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